

COURSE BRANCH

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1. Introduction

1.1 purpose: The purpose of the system is to ease creation of schedules for students in the greater Boston area (BU,Northeastern,UVM) with planning software to design a visualizing tree. Currently there is no strong way to visualize the course selection and planning process.

1.2 Scope:

- a) Course Branch
- b) Students at various universities often struggle with course selection, due to a lack of effective ways to visualize prerequisites, understand degree requirements and view their potential course paths. Current university scheduling portals lack the intuitive design and visualization features, while providing little to no information regarding difficulty and quality of each course. [-CourseBranch addresses these issues by offering a visual tree-based planning tool that will integrate with course requirements, ratings, and scheduling options. It will support exploration, planning, and comparison of degree branches, while leaving official registration functions to the existing university systems.
- c) During the creation of schedules for students there is currently no way to visualize what classes lead into other classes. Our program will help to visualize the requirements, differences in classes and courses, quality professors, example course paths of previous students, and updated course options at different schools. The courses that we provide will also satisfy all college specific requirements (such as BU's hub credits).

1.3.1 Product Perspective

- a) CourseBranch is a standalone tool, that will be used for planning and reviewing schedules. It is not meant as a replacement for university scheduling systems, but rather a complement to provide students with a better system to research, compare and plan degree paths. It will integrate with a number of other products:
 - i) University and Scheduling Portals

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- 1) CourseBranch will work alongside official university portals. Students will be able to plan and export their schedules from CourseBranch, and then register the course through official channels.
- ii) Public Data Sources
 - 1) CourseBranch will integrate with various college websites, as well as RMP, and other sources we may deem viable. These sources will be used to import crucial information like course descriptions, degree requirements, and professor ratings.
- iii) Auth Systems
 - 1) CourseBranch will also need to include authentication with the school email system, in order to provide students with secure access to the platform.

1.3.2 Product Functions

1. Course Search
 - a. Students can search and browse courses by name, code, major or professor
2. Course Rating
 - a. Students can rate courses on a rating scale, and submit written feedback for each course. Other users can see the feedback and the aggregate score.
3. Degree Planning
 - a. The system is able to fetch and provide degree requirements, like necessary courses to take, minimal credits, and any other grad requirements related to scheduling. It shall then produce accurate maps of supported majors.
4. Schedule Building
 - a. Students can use the system to build their schedule by selecting the course they wish to take, in sequence, until they complete their degree. Users can assemble completed schedules (branches) and export them.
5. User Authentication
 - a. The system shall allow students to register and log in with a school email. It should provide secure login, session management and have account persistence.

1.3.3 User Characteristics

Users will be either prospective college students or current college students. As such we will require .edu emails to ensure that students are the ones leaving reviews. At the same time emails will be required for accounts so that our users are able to update and follow their progression throughout their academic career. We also will be complying with WCAG 2.1 requirements for comparable accessibility requirements as the universities we are working under.

1.3.4 Limitations

- a) regulatory requirements and policies;
WCAG requirements and university requirements such as graduation requirements.
Any other possible university requirements.

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Complies with FERPA requirements.

Complies with ISO/IEC 27001

b) hardware limitations (e.g., signal timing requirements);

Runs on chrome

c) interfaces to other applications;

Desktop / mobile accessibility.

d) parallel operation;

Any one user can build one tree at a time.

e) audit functions;

If a school chooses to audit us we will allow them to double check our databases.

f) control functions;

If a school chooses to restrict us in any way that school controls what we can promote.

g) higher-order language requirements;

Database access, private/non-mutable code,

h) signal handshake protocols (e.g., XON-XOFF, ACK-NACK);

Https and any apis and university requirements.

i) quality requirements (e.g., reliability);

Available courses and requirements will be up to date.

Reasonable creation times for course paths.

j) criticality of the application;

Can only affect someone's college experience. False information may force someone to take more classes than needed.

k) safety and security considerations;

Private information such as emails and passwords requires that we have solid information security for our users.

l) physical/mental considerations:

We are providing a positive environment for students so as to minimize mental stress.

We will assume that users are able to interact with phones and computers in a reasonable form.

m) limitations that are sourced from other systems, including real-time requirements from the controlled system through interfaces.

Schools' timeframes on updating courses. A reasonable timeframe of making the visual representation (> 5 secs)

1.4 Definitions

WCAG 2.1 - Web Content Accessibility Guidelines, an international standard for digital content, that provides accessibility for people with disabilities

FERPA - Family Education Right and Privacy Act, a US law protecting student educational data and privacy

ISO/IEC 27001 - an international standard that defines the requirements for an Information Security Management System (ISMS)

XON-XOFF \ ACK-NACK - Low-level signal handshake protocols (used in serial communication).

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2. References ?

<https://www.gradescope.com/>

<https://www.getpaint.net/>

3. Requirements

3.1 Functions

Student Features:

- The system shall allow students to rate professors anonymously.
- The system shall allow students to rate courses anonymously.
- The system shall recommend courses based on prerequisites.
- The system shall provide login authentication.
- The system shall allow students to filter courses by ratings.
- The system shall allow students to compare multiple degree paths.
- The system shall allow future students to explore degree requirements.
- The system shall allow students to input completed courses manually.
- The system shall allow users to search for available courses.

Developer Features:

- The system shall allow admins to add implementation for new universities
- The system shall allow admins to manage the state of the application

System Features:

- The system shall parse course requirements from public APIs.
- The system shall integrate a course rating system.
- The system shall update course requirement data regularly.
- The system shall include a privacy policy describing how data is collected and used.
- The system shall encrypt user data.
- The system shall provide an expandable way to add new university catalogs

3.2 Performance Requirements

- The system shall be able to serve at least 200 users simultaneously
- The system shall be able to process at least 10 transactions per second
- The system shall allow any user to create one tree at a time
- The system shall generate the course tree graphics in less than 5 seconds

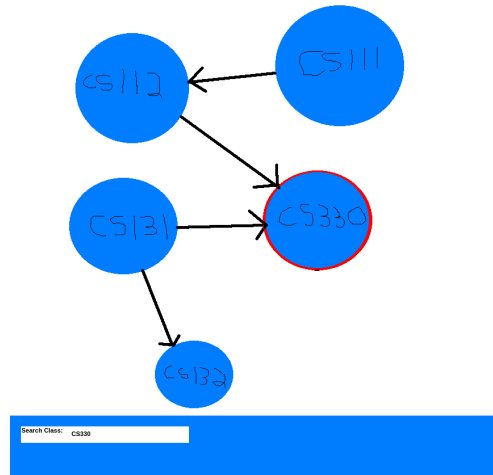
3.3 Usability Requirements

- The system shall show more relationship information at its highest level of abstraction than BU's course catalogue
- Shall not show outdated information to the customer about their courses, as to prevent any misunderstandings and inhibiting their academic career
- The system shall be accessible to WCAG 2.1 standards

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- The system shall cover over 90% of classes within the catalogue
- The system shall integrate with .edu institutional logins
- The system shall be easily to navigate with both mouse and arrow keys

Searching Class Interface



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Login Screen - Inspired by Gradescope



Select your school below

If you do not see your school in this list, that means we do not have single sign-on enabled at your school yet.

3.4 Interface Requirements

- Backend Access
 - To access low level information of database information
 - From the database to the developer
 - As accurate as current information on the database is
 - Classes per Branch
 - Instant / ASAP
 - Should be higher access than the website
 - All developer commands including mutability and unlimited reading
 - Classes, Relationships, Branches

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- Website Access
 - To access higher level pristine client facing view of database information
 - From database to website
 - As accurate as current verified information on database is
 - Classes and their relationships
 - Requirements Documentation specifies this
 - Should be lower than dev mode, with only UI-ified information being visible
 - Should have read only access
 - Classes, Relationships, Branches

3.5 Logical Database Requirements

- Course Trees for each school
 - Every use of the website
 - Through an internet connection with an account
 - Classes and their relationships to each other
 - Public information, just don't manipulate it in base content to display false information
 - We can retain data for as long as needed because its public university information
- Course Ratings
 - With every use of the website, often semesterly due to rating classes after they've finished
 - Through internet connection with an account
 - Classes and their ratings provided by users
 - We will not alter the reviews but will censor any inappropriate language, and will display every review fitting this criteria
 - Whatever GDPR says, we will present those to the user at time of review creation

3.6 Design Constraints

- The system shall comply with FERPA by not exposing private data
- The system shall comply with accessibility standards (WCAG 2.1)
- The system shall support mobile devices
- The system shall comply with ISO/IEC 27001

3.7 Software System Attributes

Security:

- We will adhere to the ISO/IEC 27001 standard in order to achieve the best security possible

Portability (Scalability):

- For the server, we will create a docker image, so it would be portable for all systems and it would be extremely scalable
- For the client, the app will be a web app, which would make it usable across all operating systems
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3.8 Additional Information

- It's important to acknowledge that every single business and data decision completely and entirely relies on the cooperation from the universities and as such we will comply with every standard that they provide and every request they give
- We will attempt to make the code as optimized as we can as to minimize electricity and storage cost

4. Verification

We will ensure to include comprehensive testing of edge cases and random fuzzing for all critical information paths, with at least 1 test per function

We will manually scrape through the database to ensure correctness of information through random selections of courses for audit

We will extensively test and document pretending to be users of the website as to explore all use cases and find any common surface level bugs

We will reason about data paths and execution paths to try and find any logical or timing errors such as deadlocks, null pointer exceptions, etc.

We will use linting software to find any surface level syntax errors

5. Appendices

5.1 Assumptions and dependencies

- The project depends on and assumes the public availability of university course data
- The project assumes that most operating systems have a working web browser made for them
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5.2 Acronyms and Abbreviations

WCAG 2.1 - Web Content Accessibility Guidelines, an international standard for digital content, that provides accessibility for people with disabilities

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